

Factory-to-Customer Shipping Route Efficiency Analysis for Nassau Candy Distributor

Project Research Paper • Data Analytics and Decision Support

Abstract

This project presents a data-driven framework for analyzing factory-to-customer shipping route efficiency for Nassau Candy Distributor. The objective is to transform shipment data into practical insights about route performance, geographic bottlenecks, shipping modes, data quality, and operational decision-making. A Streamlit dashboard was developed to make the analysis interactive and accessible. The application combines an executive KPI overview with data-quality auditing, route efficiency ranking, geographic bottleneck analysis, ship-mode comparison, route drill-down, and decision-support recommendations.

1. Introduction

Efficient transportation is an important component of supply-chain performance. Shipment-level data can reveal differences in route behavior, geographic concentration, shipping modes, and potential data-quality problems. This project addresses the challenge of turning a shipment dataset into an interactive analytical tool for factory-to-customer shipping analysis.

2. Objectives

The objectives are to analyze shipment and route information; provide an executive KPI overview; audit data quality; rank routes; identify geographic bottlenecks; compare shipping modes; enable route-level drill-down; and translate analytical findings into practical recommendations.

3. Dataset and Data Preparation

The primary dataset is Nassau Candy Distributor.csv. Data preparation is treated as an explicit stage of the workflow. Relevant shipment, route, geographic, and shipping-mode fields are examined. Questionable or incomplete values are intended to be surfaced as data-quality warnings rather than silently changed, preserving transparency in the analysis.

4. Methodology

The dashboard is organized into seven modules: (1) Executive KPI overview, (2) Data quality audit, (3) Route efficiency ranking, (4) Geographic bottleneck analysis, (5) Ship-mode comparison, (6) Route drill-down, and (7) Decision-support recommendations. This layered workflow allows a user to move from overall performance to route-specific investigation.

5. System Implementation

The solution is implemented as a Streamlit web application. The main application is app.py, dependencies are listed in requirements.txt, and the Nassau Candy Distributor CSV is kept with the application files. The project is version-controlled in GitHub and deployed as a Streamlit app.

6. Results and Discussion

The dashboard provides a unified environment for examining route efficiency rather than relying on a single static metric. Users can review KPIs, validate data quality, compare routes and shipping modes, investigate geographic patterns, and drill down into individual routes. This design supports both management-level review and detailed analytical investigation.

7. Decision-Support Recommendations

Routes that repeatedly appear inefficient should be investigated further. Geographic bottlenecks should be examined before changing transportation decisions. Ship-mode comparisons can support evidence-based choices. Data-quality warnings should be reviewed before high-impact decisions, and the same indicators can be monitored periodically to evaluate improvement.

8. Limitations

The analysis depends on the accuracy, completeness, and time coverage of the provided dataset. Real-world factors such as carrier contracts, service-level commitments, traffic, fuel prices, inventory priorities, and customer-specific constraints may not be fully represented. Therefore, the dashboard should be treated as a decision-support tool and its findings validated operationally.

9. Conclusion

This project demonstrates how shipment data can be transformed into an interactive route-efficiency and decision-support system. Combining data-quality auditing, KPI analysis, route ranking, geographic investigation, shipping-mode comparison, and drill-down analysis provides a practical framework for understanding factory-to-customer shipping performance.

Project Resources

GitHub repository: github.com/KSHITIZ-HUB/nassau-candy-shipping-efficiency

Deployed application: 9nbfqy2ec573fmffi99ttp.streamlit.app